

fabricated anywhere. For most designs, a little thought and planning will save cost and improve manufacturability.”

Creating a proper digital representation of the circuit before taking it to production is also important. The PCB Power team continues, “Define all the components and circuit schematic properly in the CAD software during design. In case you have missed out a component, define the component in your CAD library in the correct way and process all the information from there.”

Ensure protection against electrostatic discharge. Electrostatic discharge (ESD) is caused by sudden electrical energy, developed by static electricity or electrostatic induction, flowing from one charged object to another. Electronic circuit components are highly sensitive to ESD. Manufacturing floors and design areas should invest in ESD-protected areas (EPAs) where high-charge objects are kept out.

Ionisation is a popular process for ESD-proofing of electronic devices that cannot be grounded. Use of ESD protector materials during circuit design becomes imperative. Some commonly used ESD protectors include metal-oxide varistors, transient-voltage-suppression diodes, polymer-based ESD suppressors and clamping diodes. ESD protection will keep electronic devices from getting damaged by heavy lightning or any kind of charge transfer.

Save the circuit from EMI. Electromagnetic compatibility (EMC) is crucial to ensure the quality of the circuit being designed. It ensures that electronic circuits are not troubled by electromagnetic interference (EMI).

EMI arises due to radio-emission of other nearby electronic objects. It impacts electronic circuits by providing wrong data or making the system stop altogether. EMI shielding can be done by using components like ferrite beads and capacitor filters. Overall, EMC can be controlled using shielded inductors to cut down sharp current transients. There are also EMI shielding materials that can keep the PCB unaffected.

Create resilience to reverse polarity. It always has the chance that the two charged ends of an electronic device get connected to the wrong terminals. One of the most common exam-



Fig. 1: ESD testing in an EPA

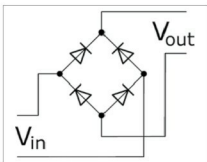


Fig. 2: A diode bridge rectifier circuit

ples is attaching a battery in reverse polarity to an electronic device. If the circuit is not well-protected against reverse polarity, hazardous outcomes might follow, including malfunction or even frying of the circuit. One precautionary measure is to connect the heavy-load-carrying microprocessor output drivers to a large resistor. Using a diode bridge is another option that rectifies the input and ensures correct polarity in the circuit.

Tend to each component separately for thermal resilience. Electronic circuits can lead to disasters if their components are not tested for heat resistivity.

The process to ensure thermal stability at peak temperature differs for every component. For instance, a resistor's capacity is determined by measuring the de-rating of power at peak temperature. Capacitance, on the other hand, can be checked by evaluating the leakage rate, increase in equivalent series resistance capacity and de-rating. For diodes and transistors, current gain and recovery time under peak temperature can be essential determinants.

Evaluate the circuit behaviour with power. It is essential to know the behaviour of the circuit when it is connected to a power source. Measuring the pull-up resistors can help you know whether the circuit is

behaving properly during power-up. Pull-up resistance on a logic circuit's microcontroller units should be high on power-up.

You should also know how your circuit will function if there is a sudden outage in power while it is processing some data. Including a power-down detection circuit is very important to ensure that the processor is saving the information properly during a sudden power cut. Power-off detection circuit is a network of a diode and a collection of capacitors, transistors and a voltage-generating circuit that keeps a circuit running for some time with its discharged power during power-cut.

Also take precautionary measures for current overshoots and undershoots in the circuit. These are basically spikes and falls in the current flowing through a circuit. Very high overshoots may fry the circuit. On the other hand, excessive falls can hamper the circuit functioning. Use of clamping diodes is a good option to ensure safety of circuit terminals from such power inconsistencies.

Test points are important. Test points are small wired loops placed on a PCB board to allow easy anchoring of test probes on the circuit. Usually, test probes are made of phosphor-bronze alloy with copper conductors or silver-plate coatings. Ground nets, power nets and threshold points of circuits should be mandatorily assigned test points.

Be clear with your manufacturer. As mentioned before, clarity with the manufacturer right from the start is very essential. To avoid complications, learn about the manufacturer's specifications including minimum trace width, spacing and number of layers. You must have complete transparency while sharing the design details. Mention all the components, the schematic and the build. In case a custom-build circuit is desired, but has not been communicated to the manufacturer, they would be following the default build model, which might create great differences from the expected outcome.

Electronic circuit design is not easy. Nevertheless, if done well, it can do wonders—whether you are doing a small-scale project or a large-scale business. Thus, keeping the rulebook open will keep you a step ahead in the game. **RFV**